Name: Thabo Aphane

Student number: 57709408

Module: SME 3701

Assignment: 02

Question 1:

1.1

$k_t = \frac{GJ_p}{L}$

$\therefore J_p = \frac{\pi d^4}{32}$

$= \frac{\pi(0.050)^4}{32}$

$= 6.136x10^{-7}\ m^4$

$\therefore k_t = \frac{(79.3x10^9)(6.136x10^{-7})}{1}$

= 486578.70 Nm/rad

$J_o\ddot{\theta} + K_t\theta = M_t(t)$

$\therefore T = \frac{60}{1000} = 0.06\ s$

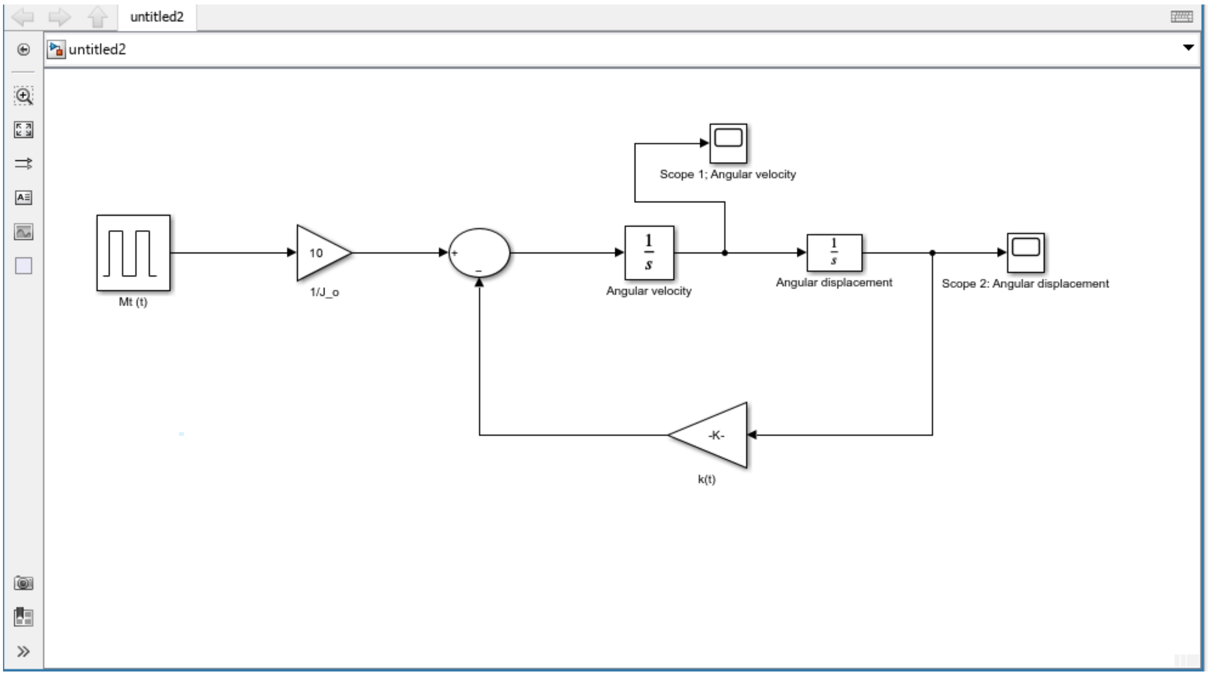
$\ddot{\theta} = \frac{1}{J_o}M_t(t) - \frac{K_t\theta}{J_o}$

= $10\ M_t(t) - 486578700\theta$

∴Pulse width: $\frac{1\ broken\ tooth}{16\ total\ teeth}x100 = 6.25\%$

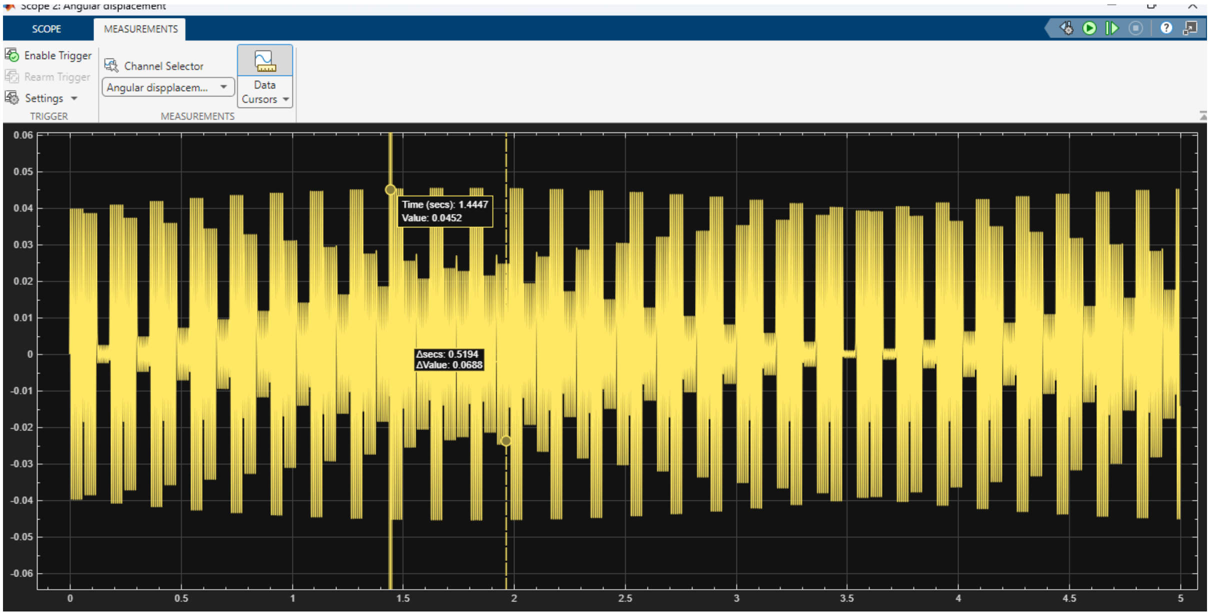
Parameter	Value	Unit
Driven gear inertia J_o	0.10	$kg.m^2$
Shaft diameter d	0.050	m
Shaft length L	1.0	m
Shear modulus G	79.3	GPa
Driving gear speed	1000	rpm
Transmitted torque	1000	N.m
Number of teeth	16	-
Broken teeth	1	-
Pulse width	6.25	%

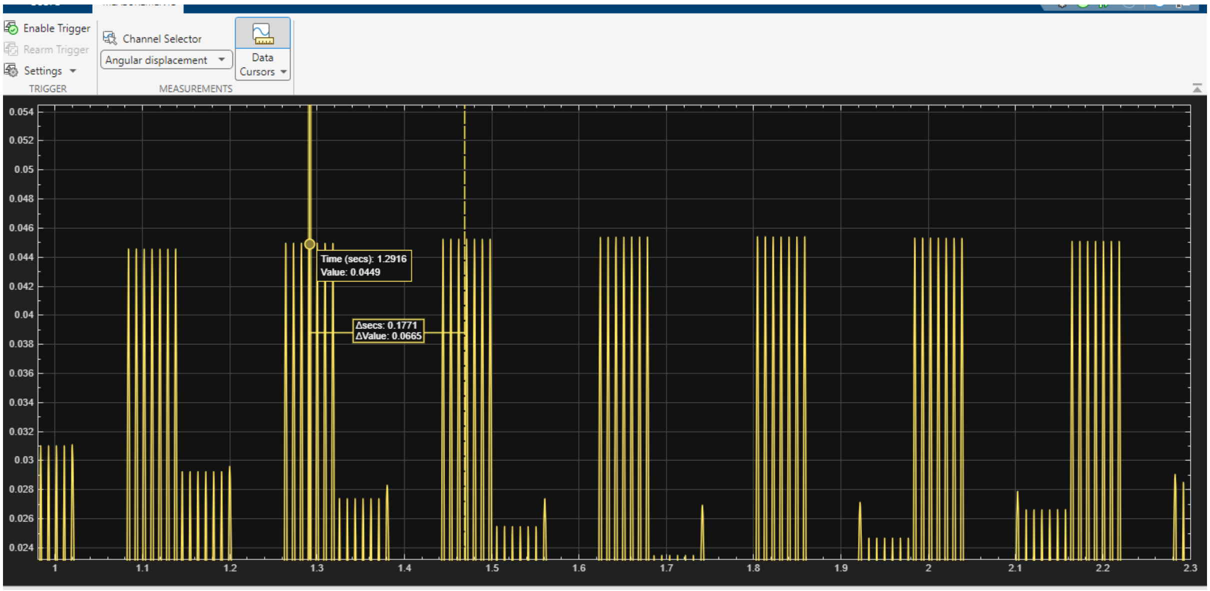
Simulink model of the geared shaft system



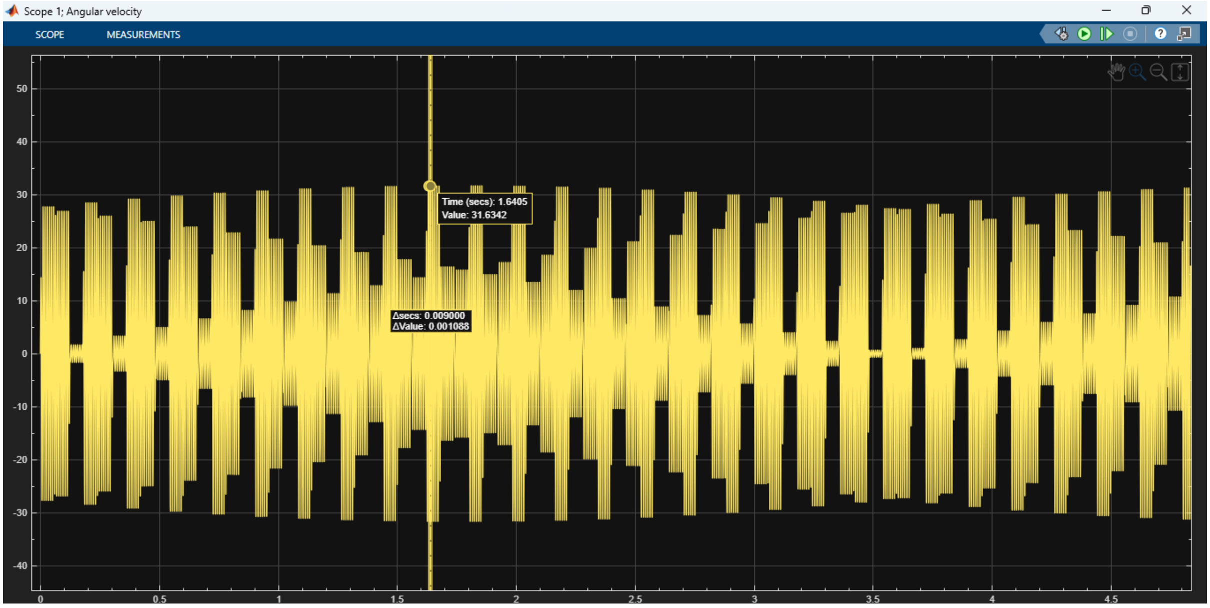
1.2

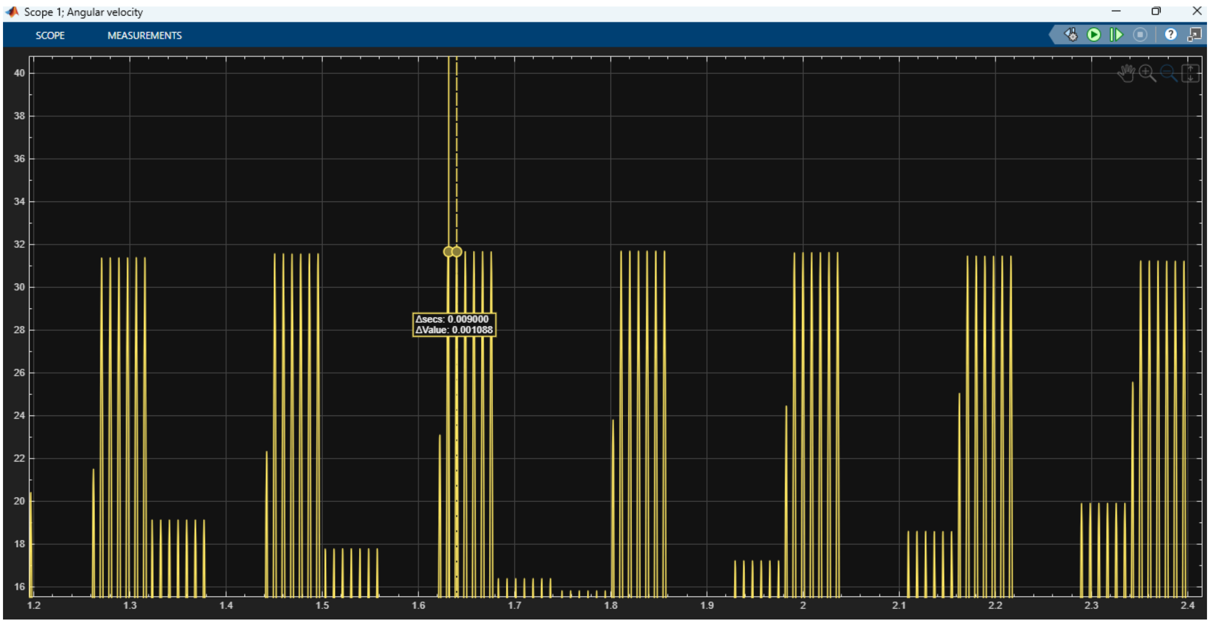
Angular Displacement:





Angular velocity:





1.3

From the Simulink simulations:

- Maximum angular displacement = 0.0049 rad
- Maximum angular velocity = 31.6353 rad/s

1.4

$\mathrm{T} = 0.00900\ \mathrm{s}$

∴ The dominant torsional vibration frequency of the driven gear is:

$f = \frac{1}{T}$

$= \frac{1}{0.00900}$

$f = 111.11\ Hz$

Because there are 16 teeth, a broken tooth means the torque drops and recovers once per revolution.

The response is oscillatory with repeated disturbances, which is what we expect from the periodic broken-tooth excitation.

The response shows sustained periodic oscillations rather than a decaying transient response. This indicates that the system is experiencing resonant or near-resonant torsional vibration, caused by the periodic torque disturbance associated with the broken gear tooth.